Session Objectives

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1. Perform a focused Osteopathic Structural Exam relevant to the pulmonary system including TART changes, segmental diagnoses, assessment of the rib cage (including the sternum, ribs, thoracic inlet, thoracic diaphragm), and upper extremity somatic dysfunctions.
2. Perform Osteopathic Manipulative Treatments to relevant and pertinent somatic dysfunctions using the techniques taught and demonstrated in this lab as well as previous labs.
3. Consider how each of the Five Models of Osteopathic Care: Biomechanical Model (OCMM and Fascia), Respiratory-Circulatory Model (Lymphatics), Neurological Model (Autonomic Nervous System), Metabolic Model (Inflammation, Homeostasis, Nutrition and Exercise), and Behavioral Model (Life Style Modification, Stress Management, and Environment) may be applied in the context of each of the techniques taught in lab.

Pulmonary System Consideration

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Region	Diagnosis	Treatment
• Cranial	• Vagus nerve: Somatic dysfunction pertaining to fascial restriction around the nerve as it passes through the jugular foramen. Overstimulation leads to bronchoconstriction and increased secretion of the bronchial glands.	• OA Decompression • Occipital Mastoid Suture Release
• Cervical	• Superior cervical sympathetic ganglia: Overstimulation leads to bronchodilation and decreased secretions • Vagus Nerve • Phrenic Nerve • Accessory muscles of respiration	• BLT AA, C1-C3 • BLT AA • BLT C3-C5, MFR Scalene muscles • MFR Sternocleidomastoid muscles
• Thoracic	• Sympathetic Chain Ganglia: normalizing sympathetic tone will improve bronchodilation, sputum production, and lymphatic flow. • Sternum: fascial restrictions will lead to decreased compliance of the rib cage and suboptimal thoracic excursion, oxygenation, and lymphatic flow.	• BLT T1-T7 • MFR Sternal Release

Pulmonary System Consideration

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Region	Diagnosis	Treatment
• Thoracic Inlet	• Pulmonary lymph passes through hilar/tracheobronchial nodes into the bronchomediastinal trunks, with right-sided drainage generally proceeding toward the right lymphatic duct and left-sided drainage toward the thoracic duct, but there is important anatomic variability and crossover	• MFR Thoracic Inlet
• Respiratory Diaphragm	• Inhalation dysfunction or exhalation dysfunction of one or multiple parts of the thoracic diaphragm restricts thoracic excursion and leads to suboptimal oxygenation, poor perfusion, and atelectasis. Treatment improves thoracic excursion which results in optimized oxygenation and draws lymphatic fluid through the thoracic duct.	• MFR Thoracic Diaphragm
• Ribs	• Inhalation or exhalation dysfunction: treatment will improve respiratory excursion biomechanically.	• BLT Ribs 1-12

Pulmonary System Consideration

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Region	Diagnosis	Treatment
• Lumbar	• Attachments of the thoracic diaphragm L1-L3. Treating this region alleviates chronic low back pain associated with asthma or COPD due to somatic dysfunction in the diaphragm. Restriction here may also lead to suboptimal lymphatic flow through the cisterna chyli past the crura of the diaphragm.	• BLT L1-L3
• Pelvic Diaphragm	• Restriction in either hemidiaphragm leads to decreased thoracic excursion, as the lungs are prevented from extending down into the abdomen on inspiration.	• MFR of the right and left innominate, or pelvic hemi-diaphragm.
• Sacrum	• Restriction in sacral flexion or extension may restrict the pelvic diaphragm, similarly leading to decreased thoracic excursion.	• BLT Sacrum
• Upper Extremities	• Sternoclavicular Joint • Acromioclavicular Joint • Glenohumeral Joint • Scapula: Restriction in this area will decrease the thoracic excursion of the upper ribs, which leads to suboptimal lymphatic drainage.	• BLT SCJ • BLT ACJ • BLT GHJ • MFR Scapula

Articulatory Technique

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Indications

- Rib raising may be used in the entire thoracic cage
- Somatic dysfunction in the rib or thoracic regions, resulting in:
  - **decreased thoracic excursion**
  - **increased work of breathing**
  - **impaired lymphatic drainage**
  - **facilitation of one or more segments in the thoracic spine (VSR/SVR/VVR/SSR)**

Contraindications

- Fractured ribs
- Vascular compromise (e.g. secondary to fracture, chest tube placement)
- Malignancy/bony metastases to rib or vertebrae
- Superficial Infection (e.g., cellulitis, abscess)
- Bleeding disorders



Myofascial Release

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Indications

- Somatic dysfunction involving somatic or visceral myofascial tissues
- Neuroreflexive abnormalities
- Circulatory restrictions
- Lymphatic restrictions
- Bronchial restrictions
- Homeostatic imbalance

"In systemic disease, MFR techniques address structural imbalances and somatic dysfunction indicative of and pursuant to changes in homeostasis. They aid in the normalization of function in all systems."

Contraindications

Absolute:

- Absence of somatic dysfunction

Relative:

- Fractures
- Open wounds/soft tissue or bony infections
- Abscesses
- Deep venous thrombosis (threat of embolism)/anticoagulation
- Disseminated or focal neoplasm
- Recent postoperative conditions over the site of proposed treatment (wound dehiscence)
- Aortic aneurysm